

SAT UZBEKISTAN

Telegram @satuzbhunters

- 2 1. The graph of the equation below in the xy -plane is a linear function. Which of the following equivalent forms of the equation includes the x -intercept as a constant?

$$y = 2x + 6$$

- A) $y - 2x = 6$
- B) $\frac{1}{2}y - x = 3$
- C) $y = 2(x + 3)$
- D) $x = \frac{1}{2}y - 3$

- 2 2. The graph of the equation below in the xy -plane is a parabola. Which of the following equivalent forms of the equation includes the x -coordinate of the vertex as a constant?

$$y = x^2 - 8x + 12$$

- A) $y = (x - 4)^2 - 4$
- B) $y = (x - 6)(x - 2)$
- C) $y = x(x - 8) + 12$
- D) $y - 12 = x(x - 8)$

- 2 3. The graph of the equation below in the xy -plane is a linear function. Which of the following equivalent forms of the equation includes the y -intercept as a constant?

$$2x + 3y = 12$$

- A) $x = -\frac{3}{2}y + 6$
- B) $3y = -2x + 12$
- C) $\frac{2}{3}x - 4 + y = 0$
- D) $y = -\frac{2}{3}x + 4$

- 2 4. If $f(x) = a(x - b)(x + c)$, where a , b , and c are constants, which of the following lists the x -intercept(s) of $f(x)$?

- A) $-ab$ and ac
- B) a , $-b$, and c
- C) b and $-c$
- D) $-b$ and c

- 2 5. The graph of the equation below in the xy -plane is a linear function. Which of the following is a form of the equation that includes the x -intercept as a constant?

$$6x + 4y = 24$$

- A) $y = \frac{3}{2}x + 6$
- B) $y = -\frac{3}{2}x + 6$
- C) $x = -\frac{2}{3}y + 4$
- D) $x = -\frac{2}{3}y + 6$

- 2 6. If $y = 5x^2 - 16x + 3$ is graphed in the xy -plane, which of the following elements of the graph is displayed as a constant?

- A) x -intercept
- B) y -intercept
- C) x -coordinate of the vertex
- D) y -coordinate of the vertex

- 2 7. The equation below represents a parabola in the xy -plane. Which of the following equations displays the x -intercepts of the parabola as constants or coefficients?

$$y = x^2 - 13x + 36$$

- A) $y = (x - 4)(x - 9)$
- B) $y = x(x - 13) + 36$
- C) $y = x^2 - 13(x + 3) - 3$
- D) $y - 36 = x^2 - 13x$

- 3 8. The equation below is quadratic. Which of the following equivalent equations displays the minimum value of y as a constant or coefficient?

$$y + 12x = x^2 + 9$$

- A) $y = x^2 - 12x + 9$
- B) $y = (x - 6)^2 - 27$
- C) $y = x^2 - 2(6x - 4.5)$
- D) $y = -x(x + 12) + 9$

Telegram @satuzbhunters

9. The equation below represents a parabola in the xy -plane. Which of the following is an equivalent form that displays the y -intercept of the parabola as a constant or coefficient?

2

$$y = -3(x + 2)(x + 3)$$

- A) $y = -3x^2 - 15x - 18$
B) $y = -3x^2 - 5x - 6$
C) $y = -3x^2 + 15x + 18$
D) $y = -3x - 9x + 6$

10. The equation below represents a parabola in the xy -plane. Which of the following equivalent forms of the equation displays the x -intercepts of the parabola as constants or coefficients?

2

$$y = x^2 - 6x + 8$$

- A) $y - 8 = x^2 - 6x$
B) $y + 1 = (x - 3)^2$
C) $y = (x - 4)(x - 2)$
D) $y = x(x - 6) + 8$

11. Parabola X in the xy -plane has the equation of $y = x^2 - z$. Which of the following is equal to the x -intercept(s) of the parabola as constants or coefficients?

3

- A) $z, -z$
B) $-z$
C) z^2
D) $\sqrt{z}, -\sqrt{z}$

12. The function f is defined as $f(x) = a(1 + b)^x$, where a and b are positive constants and $x \geq 0$, which of the following best describes the constant a ?

3

- A) The maximum of the function.
B) The x -intercept.
C) The minimum of the function.
D) The rate of increase of the function.

13. The function f is defined by the equation below. In which of the following equivalent forms of the equation does the maximum value of $f(x)$ appear as a constant or coefficient?

3

$$f(x) = -3x^2 + 24x - 32$$

- A) $f(x) = -3x^2 + 18x + 6x - 36$
B) $f(x) = -3x(x - 6) + 6(x - 6)$
C) $f(x) = -3(x - 4)^2 + 16$
D) $f(x) = -3(x - 2)(x - 6)$

14. Parabola A in the xy -plane has the equation $x - y^2 + 8y + 5 = 2y + 13$. Which of the following is an equivalent form of the equation that shows the y -intercept(s) of the parabola as constants or coefficients?

3

- A) $x = y^2 - 6y + 8$
B) $x = y^2 - 6y + 18$
C) $x = (y - 4)(y - 2)$
D) $x = (y - 3)^2 - 1$

15. If $f(x) - d = a(x - b)^2 + c$, where a, b, c , and d are constants, which of the following is equal to the y -coordinate of the vertex of $f(x)$?

4

- A) b
B) $c + d$
C) $-ab$
D) c

16. Parabola Q in the coordinate plane has the equation $x - 3y^2 - 18y + 13 = 0$. Which equation shows the x -intercept(s) of the parabola as constants or coefficients?

3

- A) $x = 3(y - 3)^2 - 14$
B) $x = 3y^2 + 18y - 13$
C) $x + 14 = 3(y - 3)^2$
D) $y = \sqrt{\frac{x+14}{3}} + 3$

17. If $(y - 3)^2 + 8 - 2x = 0$ is graphed in the xy -plane, which of the following elements of the graph is displayed as a constant?

4

- A) x -intercept
B) y -intercept
C) x -coordinate of the vertex
D) y -coordinate of the vertex

Telegram @satuzbhunters

18. The functions f and g are defined by the given equations, where $x \geq 0$. Which of the following equations displays, as a constant or coefficient, the minimum value of the function it defines, where $x \geq 0$?
- I. $f(x) = 270(0.7)^x$
 II. $g(x) = 270(1.7)^x$
- A) I only
 B) II only
 C) I and II
 D) Neither I nor II
19. Parabola B in the xy -plane has the equation of $y = x^2 - ax - bx + ab$. Which of the following is equivalent to the x -coordinate for the vertex of the parabola?
- A) $\frac{a+b}{2}$
 B) $\frac{-a-b}{2}$
 C) ab
 D) $\frac{a}{b}$
20. Which of the following equivalent forms of the given function below displays the y -coordinate of the y -intercept of its graph in the xy -plane as a constant?
- $$f(x) = 9^{-(2+x)}$$
- A) $f(x) = \frac{1}{81} \left(\frac{1}{9}\right)^x$
 B) $f(x) = \left(\frac{1}{9}\right)^{(2+x)}$
 C) $f(x) = 3^{-2(2+x)}$
 D) $f(x) = 9^{(-2-x)}$
21. For the exponential function f , the value of $f(1)$ is a , where a is a constant. Which of the following equivalent forms of the function f shows the value of a as the coefficient or the base?
- A) $f(x) = 37(2)^{x+1}$
 B) $f(x) = 70(2)^{x-1}$
 C) $f(x) = 135(2)^{x+2}$
 D) $f(x) = 235(2)^{x-2}$
22. The functions f and g are defined by the given equations, where $x \geq 0$. Which of the following equations displays, as a constant or coefficient, the maximum value of the function it defines, where $x \geq 0$?
- I. $f(x) = 18(0.6)^{x+1}$
 II. $g(x) = 18(0.6)(0.6)^{x-1}$
- A) I only
 B) II only
 C) I and II
 D) Neither I nor II
23. Which of the following equivalent forms of the given function f displays, as the base or the coefficient, the y -coordinate of the y -intercept of the graph of $y = f(x)$ in the xy -plane?
- $$f(x) = 4^{-2(x+1)}$$
- A) $f(x) = \left(\frac{1}{4}\right)^{(2x+2)}$
 B) $f(x) = \left(\frac{1}{16}\right) \left(\frac{1}{4}\right)^{2x}$
 C) $f(x) = \left(\frac{1}{81}\right)^{\left(-\frac{1}{2}x - \frac{1}{2}\right)}$
 D) $f(x) = 4^{-2x-2}$
24. The functions f and g are defined by the given equations, where $x \geq 0$. Which of the following equations displays, as a constant or coefficient, the minimum value of the function it defines, where $x \geq 0$?
- I. $f(x) = 103 \left(\frac{1}{1.1}\right) (1.1)^{x+1}$
 II. $g(x) = 103(1.44)(1.2)^{x-2}$
- A) I only
 B) II only
 C) I and II
 D) Neither I nor II